

PEPSICO UNLOCKS GRANULAR GROWTH USING A DATA-DRIVEN UNDERSTANDING OF SHOPPERS

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CASE STUDY

an in-depth description of a firm's approach to an IT management issue (intended for MBA and executive education)

DATA WRAPPING
DATA MONETIZATION
DATA SCIENCE
DIGITAL TRANSFORMATION
CUSTOMER CO-CREATION
PARTNERING
INFORMATION SOLUTIONS
ENTERPRISE SERVICES
DATA-DRIVEN DECISION MAKING
MARKETING

In 2019, PepsiCo was a $65 billion global consumer packaged goods company that offered a suite of data analytics capabilities called PEPWORX℠. The capabilities enabled PepsiCo and its retail customers to unlock granular growth in the marketplace by helping users of PEPWORX℠ better understand how to optimize store-level assortment and total store space, manage marketing dollars more effectively, and successfully launch and manage innovative marketing programs. PEPWORX℠ was the product of a centralized unit at PepsiCo called the PepsiCo Demand Accelerator, established in early 2015, which helped the company build an advanced analytics platform, a large anonymized shopper-level data set, and an array of information solutions, ultimately packaged as PEPWORX℠. Over that time, the PEPWORX℠ capabilities supported PepsiCo in transforming the nature of its retail customer relationships from transactional to collaborative.

MIT MANAGEMENT SLOAN SCHOOL | CENTER FOR INFORMATION SYSTEMS RESEARCH (CISR)

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In 2015, PepsiCo established a centralized unit called the PepsiCo Demand Accelerator to produce innovative new data-driven marketing services across the enterprise. Over the following four years, the Demand Accelerator helped establish an advanced analytics platform, a large anonymized shopper-level data set, and an array of information solutions to support the sales and marketing needs of PepsiCo—and of PepsiCo's retail customers. By 2019, the Demand Accelerator formally packaged its shopper insights into a suite of data analytics capabilities called PEPWORX℠. The capabilities enabled PepsiCo and its retail customers to unlock granular growth in the marketplace by helping users of PEPWORX℠ better understand how to optimize store-level assortment and total store space, manage marketing dollars more effectively, and successfully launch and manage innovative marketing programs. The capabilities also helped PepsiCo transform the nature of its retail customer relationships from transactional to collaborative.

Frito-Lay is on a mission to create more smiles with every bite. Strong customer partnerships are vital to our success. When we work together and leverage meaningful, tech-enabled insights, we are better positioned to deliver the products consumers want, where and when they want them.

STEVEN WILLIAMS, CHIEF EXECUTIVE OFFICER, FRITO-LAY NORTH AMERICA

This case study was prepared by Barbara H. Wixom of the MIT Sloan Center for Information Systems Research (CISR). The case was written for the purposes of class discussion, rather than to illustrate either effective or ineffective handling of a managerial situation. The author would like to acknowledge and thank the executives at PepsiCo for participating in the case study.
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PEPSICO'S GROWTH STRATEGY

In 2019, PepsiCo was a $65 billion global consumer packaged goods company headquartered in Purchase, NY with 267,000 employees worldwide (114,000 within the United States).[1] The company sold its products in approximately two hundred countries and generated 62 percent of its net revenues from three divisions in North America: PepsiCo Beverages North America (33 percent), Frito-Lay North America (25 percent), and Quaker Foods North America (4 percent). From the initial creation of the Pepsi formula in 1893, PepsiCo grew to become one of the world's largest suppliers of convenience foods and beverages, with brands such as Pepsi, Lay's, Doritos, Mountain Dew, Gatorade, and SunChips. But in the 2010s, it had become less straightforward for PepsiCo to achieve growth, according to Michael Lindsey, Frito-Lay Chief Transformation and Strategy Officer.

[PepsiCo has] historically been among the largest contributors to growth in the United States consumer goods space. We've been very successful. But that leads to a fundamental challenge. How do you continue to grow at the pace you've been accustomed to, and do it in a non-dilutive[2] way?

MICHAEL LINDSEY, CHIEF TRANSFORMATION AND STRATEGY OFFICER, FRITO-LAY NORTH AMERICA

In the past, the company grew in four key ways. First, PepsiCo manufactured and supplied a **large variety** of well-known products, with 22 of the company's brands each generating over $1 billion in annual retail sales. Second, the company achieved **broad reach** by placing products in hundreds of thousands of shopping locations throughout the United States. PepsiCo operated the largest fleet of trucks in the United States (ahead of companies such as UPS), and its Frito-Lay division alone employed almost 50,000 manufacturing, distribution, and sales associates in manufacturing product orders and fulfilling them directly to stores. Third, PepsiCo offered a **strong value proposition**, using leading-edge manufacturing and distribution methods that achieved economies of scale to keep the company's costs low. Fourth and finally, PepsiCo created an **appealing market presence** through promotional campaigns and consumer outreach.

PepsiCo's growth strategy came under threat in the early 2010s.

Private labels exploded. Consumers began looking for brands that were fresh and new. Variety exploded in places we didn't play.[3] There were more channels that didn't previously exist.

MICHAEL LINDSEY

Additionally, North America had become a comparatively low-growth market that was expected to remain low growth as baby boomers, a key PepsiCo demographic, aged.

In the face of these external challenges, PepsiCo leaders decided to find new ways to utilize the company's size and product variety as a competitive advantage. This would not be easy. In North America, the company offered several thousand stock keeping units, or SKUs, to the marketplace. The complete set of SKUs could not fit on a single truck nor in a single distribution center.

Company leaders believed one of the keys to success was to identify latent consumer demand and niches of demand that could collectively be large enough for the company to serve. Specifically, they wanted to use data and advanced analytics to unlock granular pockets of potential growth and inform when and where to place specific SKUs in a particular retail outlet—and to communicate precisely with the shoppers who would be interested in that product.

1 Financial information and company demographics were sourced from the PepsiCo 2018 annual report, available at https://www.pepsico.com/docs/album/annual-reports/2018-annual-report.pdf?sfvrsn=35d1d2bc_2.

2 Non-dilutive refers to ensuring that revenue growth does not come at the expense of lower profit margins. Additional revenue must be as profitable as or more profitable than the existing business.

3 As an example, PepsiCo traditionally produced products based on corn and potatoes. Lindsey explained that snack aisles increasingly included products based on new core ingredients, such as chickpeas, seaweed, beets, and kale.

At the time, leaders also recognized that large retailers were accelerating their own investment in and use of analytics tools and techniques to understand their sales data and predict future patterns of shopper behavior. Jeff Swearingen, Senior Vice President of the PepsiCo Demand Accelerator, explained that this further motivated the leaders to consider ways to build PepsiCo's in-house analytics capabilities.

We looked across the marketplace and recognized that we needed to move quickly. We realized we had pockets of strength in data analytics, but they weren't well connected. We needed to bring those together, democratize them, scale them, and get more leverage from them. Many of our retail partners were moving quickly and making significant multibillion-dollar investments in the space. We knew we needed to build our own capability to ensure we had a strategic seat at the table.

JEFF SWEARINGEN, SENIOR VICE PRESIDENT, PEPSICO DEMAND ACCELERATOR

In late 2014, Swearingen and his team envisioned an ideal future analytics capability, and they presented their vision to the PepsiCo Executive Committee. The vision was summarized in a single page describing a scenario in which PepsiCo could employ analytics-based insights about the characteristics of what they later called "the modern shopper." The vision included a data-rich platform that could offer PepsiCo's retailer partners optimal product assortments, more efficient merchandising and a more engaging in-store shopper experience—in a way that would greatly improve the company's productivity and ultimately drive sales growth. During the meeting, the team assured the committee that the vision was quite attainable.

We actually had much of the capability to do all of those things. It just wasn't connected or done at scale. Our goal was to do those two things.

JEFF SWEARINGEN

The Executive Committee found the vision for a new enterprise-level shopper insights unit compelling, and invested in establishing the unit—to be called the PepsiCo Demand Accelerator—for launch in early 2015.

PEPSICO DEMAND ACCELERATOR

The PepsiCo Demand Accelerator (the DX) was staffed with approximately 180 people drawn from units across PepsiCo, including Category Management, Shopper Insights, Space Optimization, and Shopper Marketing. The new unit reported to the Chief Commercial Officer for North America and was expected to create and deliver standardized enterprise-wide data-driven marketing services. In the past, services like category management and shopper insights had existed separately in each of PepsiCo's five major business divisions. Not only were divisional services believed to be unnecessarily redundant, but they were each serving the same set of retailers in potentially inconsistent ways.

The PepsiCo Executive Committee allocated funds to invest in DX's first-year startup costs, but the committee expected the DX to self-fund over time. Swearingen and his team understood the need to generate incremental revenue growth while capturing cost synergies for PepsiCo.

We're not a think tank. We're not a center of excellence. We're not an incubator. We're a commercial team. We have an accountability to deliver incremental sales and incremental profit to our internal constituents.

JEFF SWEARINGEN

Swearingen led the DX, and he communicated from the outset that the group was embarking on a significant transformation journey. He referred to its early members as "founders," to communicate that these people were not just a part of a new team, but pioneers in building something unique and important for PepsiCo's future success.

Swearingen established three main goals for his people: (1) drive change in internal brand and commercial strategies, (2) elevate PepsiCo's position with retailers in terms of partnership and data analytics reputation, and (3) motivate PepsiCo retail partners to make better business decisions based on a new set of shopper insights. The team was granted great leeway in finding ways to achieve the goals.

The best part of this journey was that it was wide open. That is unusual in big companies. The main outcome was to drive revenue. I told [my team], "We're going to have to change some processes. We're going to have to change the way we do things."

MICHAEL CLEAVINGER, SENIOR DIRECTOR OF DATA SCIENCE AND ADVANCED ANALYTICS, PEPSICO DEMAND ACCELERATOR

At first, the DX pursued quick wins against existing business priorities and investments, such as opportunity analysis. An early win in that space was the launch of Quaker Overnight Oats, a packaged cup of dry oatmeal designed to be steeped in milk overnight for a quick breakfast the next morning. DX associates used emerging DX data and analytics capabilities to identify 24 million US households to engage, based on available demographic data such as employment data and propensity to eat healthy foods.[4] Then they helped direct marketing for the new product towards stores in locations with a higher proportion of interested households, using promotional material designed to appeal to that audience. The approach was credited for 80 percent of Quaker Overnight Oats' sales growth in the first 12 weeks after launch.

Concurrent with delivering quick wins, DX associates pursued transformational projects opportunistically.

We were pitching to anybody internally who may have challenging problems that we think we can fix. We tried to talk to a lot of people, understand what questions they had, and offer to help.

MICHAEL CLEAVINGER

They described themselves in the early days as "having tenacity," and their stakeholders agreed.

The great thing about Jeff and his team is they were constantly hungry to evolve the suite of tools to meet business needs. They were constantly on the lookout for how to evolve the tools and stay one step ahead.

JENNIFER SAENZ, PRESIDENT, PEPSICO GLOBAL FOODS

To focus their resources, DX associates looked for important, solvable problems that would lead to solutions that unlocked sales growth and gave PepsiCo a competitive advantage. Also, they believed that solutions involving PepsiCo retail partners needed to inspire retailers to act.

If our solutions produce interesting knowledge, but retailers won't (1) act on it or (2) give us credit through tangible revenue growth, then we're not going invest our precious time on it.

EVAN SHAVER, VICE PRESIDENT, SHOPPER ANALYTICS AND INSIGHTS, PEPSICO DEMAND ACCELERATOR

Swearingen structured the DX as a set of units that performed diverse but complementary activities required to deliver data-driven marketing services; having started with four units, a fifth was added in 2019 (see exhibit 1). All five units contributed to a variety of efforts such as identifying shoppers and stores, creating and delivering marketing content, and planning and executing product assortment strategies. The DX units worked closely with Marketing, Sales, and other business units across PepsiCo. The multidisciplinary nature of the DX activities required strong, seasoned leadership, and a number of associates described Swearingen as having a unique

4 The DX worked with partners whose databases identified households. PepsiCo did not maintain a database with the identities of individual households or shoppers; rather, its data about shopper behavior was anonymized.

skillset of patience, open-mindedness, and collaboration. Swearingen used these skills to engage a diverse set of internal and external stakeholders and help them stay focused and excited about delivering what was best for the shopper and the overall business.

Exhibit 1: PepsiCo Demand Accelerator Organization Chart



In 2019, PepsiCo's Demand Accelerator (DX) reported to Frito-Lay North America and PepsiCo Beverages North America. The DX initially comprised four units:

- *Shopper Marketing created programs for national promotions or for specific retailer needs.*
- *Shopper Analytics and Insights observed shopper behavior and leveraged advanced analytics to more effectively engage shoppers.*
- *Category Leadership worked with internal and external stakeholders to drive product demand and shopper loyalty.*
- *Space Optimization and Transformation developed store-level assortment and space recommendations.*

In 2019, a fifth unit, Media and Data, was added to develop a capability to connect with shoppers on a more personalized basis.

Source: PepsiCo Demand Accelerator, December 2019.

ESTABLISHMENT OF DATA AND ADVANCED ANALYTICS CAPABILITIES

In the months after the DX was established, DX associates grappled with the technical requirements for the data analytics activities they wanted to perform. The team needed PepsiCo product data standardized and aggregated across divisions. They needed much more reliable and available data about shoppers, some of it highly unstructured, from external sources. They needed the ability to process larger amounts of more diverse data to support advanced analytics techniques, including artificial intelligence.

We had to change the way we were getting the data. We realized very quickly that we couldn't rely on the technology we had right then.

MICHAEL CLEAVINGER, SENIOR DIRECTOR OF DATA SCIENCE AND ADVANCED ANALYTICS, PEPSICO DEMAND ACCELERATOR

Two significant PepsiCo IT efforts helped address the DX's technical needs. First, PepsiCo Information Management had been advancing a series of enterprise data efforts for some years. The first of these began in 2012 with the establishment of a data analytics taxonomy that established a "single version of the truth" across the 37 countries that generated close to 90 percent of PepsiCo's revenue.

This helped PepsiCo calculate brand-level profit and loss elements like cost of goods sold and trade spend, even when brands took on different names and forms around the world. PepsiCo Information Management kicked off the effort by building a product hierarchy, and then expanding that into master data management and finally into an enterprise data warehouse. By the time DX formed, the Information Management leaders were increasingly getting requests to support more advanced analytics use cases that pushed the limits of what was in place, and they were considering the creation of a new data lake to complement the data warehouse.

The enterprise data warehouse is used for very clear use patterns. It holds industrialized, curated, trusted data. As demand for advanced analytics grew, we realized we couldn't quickly do exploration or true advanced or predictive analytics using the warehouse. We were purposefully slow to embark on a data lake. We didn't want it to become a data swamp. We made sure we had a strong use case that really needed this data lake.

MARTHA ROOS, VICE PRESIDENT, IT, DIGITAL SERVICES DELIVERY, PEPSICO

The second IT effort that benefited the DX was a push by PepsiCo Enterprise Architecture to build an enhanced digital services capability. The Enterprise Architecture team looked for "anchor projects"—initiatives that required new digital capabilities and could produce value from them—to create momentum for investments. Leaders from the DX signed the initiative up to be an anchor project and to support the digital services build-out.

Together the DX and PepsiCo IT established what they called the "Next-Generation Advanced Analytics Platform" (see exhibit 2). The cloud-based platform made it possible to source and manage larger and more diverse kinds of data drawn from both inside and outside of PepsiCo. Because the platform was cloud based, it also offered secure access by retail partners as needed. The platform was rolled out in six months by using proof-of-concept prototypes and heavy involvement from DX associates. The groups carefully planned incremental deployments of capabilities so that DX associates could begin creating value even as the platform evolved.

Exhibit 2: PepsiCo's Next-Generation Advanced Analytics Platform

PepsiCo's Next-Generation Advanced Analytics Platform (NGAA) was a cloud-based ecosystem where data was gathered and ingested from on-premise and field locations. Insights were generated through machine learning and other forms of advanced analytics, and fed back for analysis or action.

Source: PepsiCo, December 2019.

Providing Most Valuable Shopper Data

The Next-Generation Advanced Analytics Platform provided access to what became core to DX activities: "Most Valuable Shopper" data. The platform contained a 20-terabyte database with 106 million records of available shopper data and more than 1,800 attributes per record. Within the database, households were scored with product probability scores (e.g., the likelihood of the household to buy a category or brand). The DX amassed the data from a variety of external and internal sources—US census data, anonymized shopper-level profiles, weather data, and product shipment—to build a rich array of demographic, behavioral, and psychographic attributes. The DX also developed information about the "shopper circuit," which represented a broader view of a shopper's spending behavior, across different retailers.

We needed to understand people, inspire people, and value people. And if you look at our data journey, a lot of it has been trying to understand people—what makes them tick, what's important to them, what are their consumption habits, etc.

GREG LYONS, CHIEF MARKETING OFFICER, PEPSICO BEVERAGES NORTH AMERICA

The Most Valuable Shopper data helped the DX develop insights about latent shopper demand, which could inform whether a particular retailer, or even specific stores within a retailer, were underperforming with PepsiCo's portfolio or in a particular category. DX associates then could suggest changes to assortment, merchandising, marketing, or pricing to increase shopper engagement and conversion. The strength of this process was its granularity. For example, looking at shopper attributes within a few block radius of a retail location could identify unfulfilled demand opportunities that would have been previously overlooked if the analysis was based only at a broader zip code level.

We had always looked at assortment in-store and tried to optimize them for shoppers. But the level of granularity and understanding of the Most Valuable Shopper data made it much more robust. We could even differentiate between stores on opposite sides of the street. We started off doing the same things as before in a much more informed way, but as we started to see the power of granularity, we were encouraged to do things really differently.

JENNIFER SAENZ, PRESIDENT, PEPSICO GLOBAL FOODS

Moving to Machine Learning

As the Most Valuable Shopper data grew, so did the DX's need for more advanced ways to analyze the data. Michael Cleavinger joined the DX from the retail industry to lead the Shopper Analytics and Insights team, but initially it was not clear exactly what kind of analytics were required.

They brought me in from the outset to do advanced analytics. What exactly that meant they didn't really know. That was the fun part. Advanced analytics was much more than a spreadsheet, which was typically what PepsiCo analysts had been using. So I came over, and they said, "Figure out what to do. How do we make assortment better?" for example.

MICHAEL CLEAVINGER, SENIOR DIRECTOR OF DATA SCIENCE AND ADVANCED ANALYTICS, PEPSICO DEMAND ACCELERATOR

The analytics evolved into progressively more sophisticated techniques. Initially, DX associates formed hypotheses using business intelligence reporting and visual representations. Then they moved into more detailed statistical modeling, which enabled the development of predictive analysis capabilities. Ultimately, the team moved to using machine learning as a key technology and methodology to create sophisticated predictive models based on the full set of DX data.

Some of the more complex retail and category problems pushed the DX into machine learning techniques. For example, one convenience store and gas (C&G) retailer wanted to maximize its fountain drink sales by store, some of which were generated by PepsiCo's brands. The retailer did not know how many people actually purchased which drinks because their scan data only reflected the purchase of a cup, not its contents.

First, DX associates combined PepsiCo's data about the number of gallons of syrup the retailer used with the retailer's data about its syrup purchases from other providers.[5] Then, the DX team developed an ensemble of machine learning models that identified specific store and shopper attributes that influenced syrup usage. The associates used this information to create ranked lists for several different C&G store profiles; each list communi-

5 When retailers provided data for projects specific to their needs, the DX did not share that data beyond the project team. This practice was important for preserving DX trust and credibility by retailers.

cated to the retailer which syrups were most important for each specific profile. The customer implemented the recommendations in late 2019 in over 1,000 stores and began measuring the impact of the recommendations to gauge next steps.

As the DX team applied more advanced data analytics tools and approaches, their non-DX PepsiCo colleagues were learning and growing their skillsets—and more deeply appreciating the possibilities afforded by data-based approaches. The stakeholders began to value new techniques like root cause analysis, trend lines, z-scores, and the Six Sigma incremental practice of asking "why" to reach an understanding of the underlying question and its answer.

Most people hate statistics, except when you make it real to them. Then they realize that they actually love statistics because it makes their job easier. We would walk people through the process and explain why and how we got to the result. We started introducing z-scores. And when they asked "why is it better to look at it this way?" we would explain that they could have a better view, not only of average sales, but of the peaks and valleys.

MICHAEL CLEAVINGER, SENIOR DIRECTOR OF DATA SCIENCE AND ADVANCED ANALYTICS, PEPSICO DEMAND ACCELERATOR

ESTABLISHMENT OF DATA AND ADVANCED ANALYTICS SOLUTIONS

Initially, the DX worked primarily with PepsiCo's Marketing and Sales teams, sometimes acting as internal consultants by gathering requirements and conducting custom projects, and at other times serving in dedicated matrixed roles. In both cases, DX associates worked closely and collaboratively with their internal colleagues—and they believed that delivering successful programs internally was critical for sustained growth and support for the DX over time.

Those internal stakeholders who can be advocates for us and our work are incredibly important in an organization of our size. It's not just us pushing our own capabilities; you've got other people who believe in them and help to create pull.

ELLEN WEBB, SENIOR DIRECTOR, SHOPPER INSIGHTS AND STRATEGY, PEPSICO DEMAND ACCELERATOR

Expanding Pepsi Zero Sugar

For its soda business, PepsiCo found that 10 percent of drinkers accounted for approximately 60 percent of soda sales. These "Loyals" were important for PepsiCo to understand so that associates could create effective strategies that met their needs and positively influenced product sales. In one case, DX associates supported the expansion of Pepsi Zero Sugar, a fast-growing, new product. The DX believed that by identifying and engaging with shoppers who were potential Pepsi Zero Sugar Loyals, they could help grow the product's market base. The idea of identifying and meeting the needs of a specific group of shoppers in a focused way was a shift for PepsiCo as the company had historically created broad-reaching marketing campaigns.

The DX defined strategies at both the store and shopper levels. To reach the right shoppers, DX associates leveraged Most Valuable Shopper data to identify households with high propensity for adoption, profiling the characteristics of a typical Pepsi Zero Sugar shopper. To identify the right stores, Shopper Analytics and Insights selected the key stores where such people shopped. Shopper Marketing then placed displays, distributed coupons, and conducted product sampling in those stores. The team experimented with different messaging, offers, and pricing based on the defined shopper characteristics. Ultimately, the program improved product sales, achieving household penetration goals, and increasing purchase frequency, a strong indicator of loyalty. The DX attributed the campaign's success in part to a more accurate understanding of and precise engagement with promising Pepsi Zero Sugar Loyals.

As DX associates collaborated with the internal PepsiCo units to drive customer-facing activities, they increasingly became integral participants of PepsiCo-retailer activities.

All three teams are working together. My team includes the general managers who manage the business. The DX team comprises subject matter experts who steer the capabilities. And then there are our retail partners, who we work with to solve their problems and integrate with their systems.

BRYAN SANTEE, VICE PRESIDENT, NATIONAL SALES, PEPSICO BEVERAGES NORTH AMERICA

Developing Custom Solutions

Often, large retail partners had large, complicated problems that required custom solutions. For example, one retailer owned thousands of stores in the United States. The retailer annually added many new stores and also refurbished existing stores. To achieve program success, the DX would need to design marketing programs specific for that retailer's particular needs instead of for the entire retail marketplace. Getting to store-level solutions for a particular retailer could prove even more successful, but it would require more effort.

The DX's capabilities appealed to PepsiCo's retailers, particularly because its shopper insights were derived from data that retailers often did not have visibility into, such as shopper behavior on a broader scale.

The retailers may have had a basic assortment process, but it was essentially based on what was in the store and what was being sold. It was typically not latent demand. So if you're a local store owner and you see that XYZ is doing well in your neighborhood, then you could locally bring that in. We were able to offer a data-based approach that systematically identified items that were selling outside [but in the area] of their stores.

KENT MONTGOMERY, VICE PRESIDENT AND GENERAL MANAGER, GLOBAL CUSTOMER TEAM

A key first step in agreeing to custom work was ensuring that the retailer was willing and able to make any changes that a future solution required—what the DX called "will and skill."

We began to model ourselves like a consulting firm, helping retailers understand how to solve their operating problems and demonstrating how we could help them do that. Were they truly interested in our perspective? Would they execute it if we presented it to them? It was imperative to understand the retailers who were eager and able to take action so that we could focus our efforts on them.

EVAN SHAVER, VICE PRESIDENT, SHOPPER ANALYTICS AND INSIGHTS, PEPSICO DEMAND ACCELERATOR

Many retailers granted the DX access to some of their retail data so that the DX could tailor solutions to the retailers' specific needs. DX associates used proofs of concept and testing to validate the effectiveness of their custom work. They also invested in experimentation; for example, A/B testing helped DX associates control for all other factors to identify the true impacts of a change being proposed or executed.

Creating Turnkey Applications

When DX associates believed that a custom solution could be scaled cost effectively to solve other retailers' problems, the DX converted the solution into a turnkey application, often with a user interface. For example, the Quaker Overnight Oats product launch effort began as a project with a national grocery chain to identify which stores were most likely to have customers who would be excited about the new Quaker Overnight Oats offering. At first, the DX produced a custom list of shopper types most likely to buy the product. Then, they identified what they called "Most Valuable Stores," which were stores that had the most of those potential shoppers, and thus would likely have the most demand. The potential shopper list took three weeks to generate and about four people to produce. The team then automated the process, which reduced the cycle time to three days. Ultimately, they built a user interface that delivered the capability in thirty minutes. Several hundred people used the

interface to run hundreds of analyses over the year on all kinds of products to identify Most Valuable Stores for any product launch.

Turnkey solutions became opportunities for the DX to serve to smaller retail players. In a desire to be "nationally great, but locally even better," PepsiCo was eager to reach these regional customers in an efficient manner. A senior director of Category Leadership was placed in each of four "megaregions"[6] to perform business relationship management roles for the categories and find ways for DX capabilities to serve customers that were managed by field and regional teams.

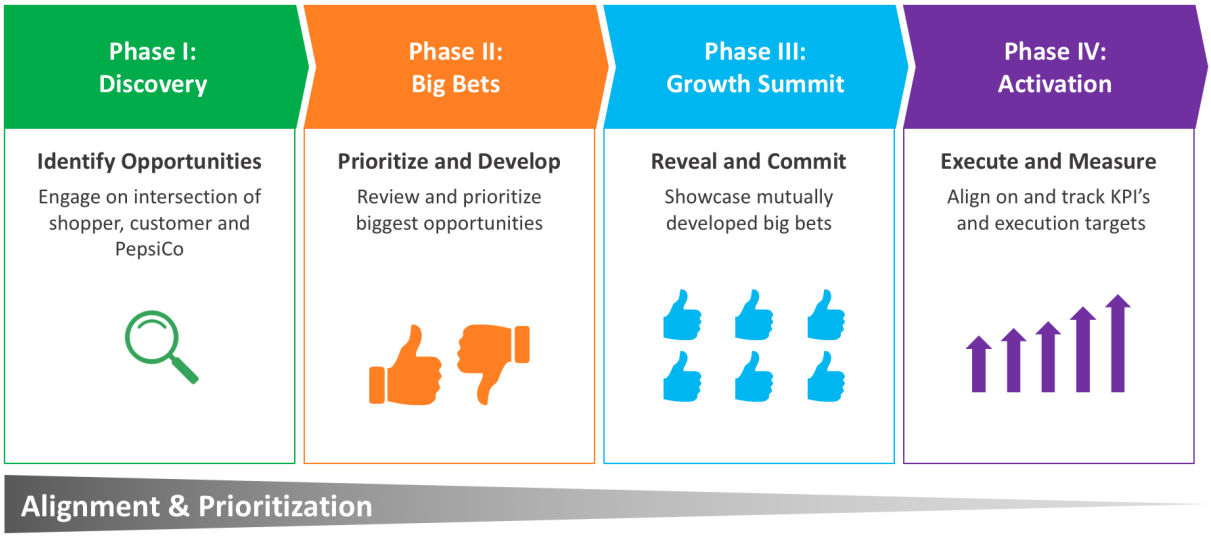
We originally focused on the biggest retailers. Over time, we knew that our regional customers could benefit, too. Serving them would take more time initially, but as we became more efficient, we could tackle these additional opportunities.

EVAN SHAVER, VICE PRESIDENT, SHOPPER ANALYTICS AND INSIGHTS, PEPSICO DEMAND ACCELERATOR

Engaging in Joint Business Planning

DX associates identified important retailer problems to solve during a formal annual process called Joint Business Planning (JBP). PepsiCo established the JBP to support the company's large partnerships so that PepsiCo and those retailers could formally plan ways to mutually drive growth aspirations (see exhibit 3). As the DX was a centralized group that crossed PepsiCo divisions, it coordinated the process across the company.

Exhibit 3: Joint Business Planning



The Joint Business Planning process, or JBP, was a formalized approach between PepsiCo and its largest retailers to annually plan activities to drive mutual growth aspirations. JBP included four key phases: Discovery, Big Bets, Growth Summit and Activation.

Source: PepsiCo Demand Accelerator, December 2019.

6 Each North America sector divided its sales regions into slightly different geographies. At the time of publishing, Frito-Lay North America had eight regions and Pepsi Beverages North America (PBNA) had four regions. To support these sectors, the DX adopted the PBNA structure and supported both sectors under a four-region model.

JBP increasingly became more interactive and iterative, which led to better collective thinking and stronger programming. PepsiCo became more focused on expanding its partners' margins and making the partners' businesses more profitable, instead of focusing solely on a PepsiCo brand gaining category market share.

We have become a more capable partner, and we play more consultative role. Over time, we have earned the trust of a lot of our key retail constituents at senior levels. They know that we're going to try to do what's best for the category. We're going to try to do what's best for our two businesses.

JEFF SWEARINGEN, SENIOR VICE PRESIDENT, PEPSICO DEMAND ACCELERATOR

DX capabilities became core to the JBP value proposition, and PepsiCo leaders attributed the capabilities with being a key influence on transforming PepsiCo retail relationships from transactional relationships to collaborative partnerships.

We use a clock as a metaphor for where you are with your customer in the relationship. 1:00 to 3:00 on that clock is a very transactional relationship. It's not deep, [and it's] very price driven. As you move around the clock, the relationship becomes more collaborative and cohesive to where you're truly trying to cocreate ways to solve problems. With the help of the DX team, we've shifted with many of these customers much farther into the collaborative side of the clock.

BRYAN SANTEE, VICE PRESIDENT, NATIONAL SALES, PEPSICO BEVERAGES NORTH AMERICA

As PepsiCo's goals became more partner oriented, its success measures moved from reflecting the progress of PepsiCo and its products to reflecting a "three-audience win," in which an effort simultaneously delivered for the shopper, the retailer, and PepsiCo. Collectively, these changes positively influenced PepsiCo's reputation as a partner.

We are confident that the customer is recognizing the value. Kantar's PoweRanking [7] is the most visible industry recognition. It has surveyed retailers for over twenty years. They've named us their #1 partner in North America for four straight years. We believe it's because retailers are seeing better solutions from us now that we're acting like one PepsiCo instead of many fragmented divisions.

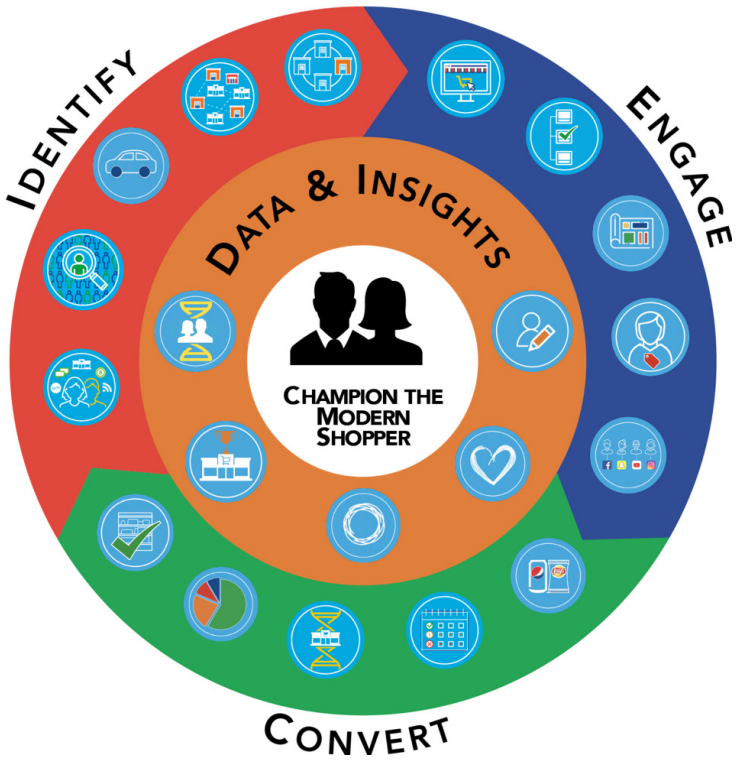
EVAN SHAVER, VICE PRESIDENT, SHOPPER ANALYTICS AND INSIGHTS, PEPSICO DEMAND ACCELERATOR

THE LAUNCH OF PEPWORX℠

By 2019, DX leaders believed that it was now possible for PepsiCo to identify latent, unmet shopper demand and to unlock granular sales growth by filling that demand. The company packaged ten of their solutions into a suite of data analytics capabilities called PEPWORX℠ (see exhibit 4). The DX offered PEPWORX℠ as a complimentary service to help PepsiCo and its retail partners personalize and tailor an assortment of products delivered to a specific retail outlet. PEPWORX℠ supported a variety of use cases—such as how to successfully launch and manage innovative marketing programs, and how to optimize total store space—that helped to differentiate PepsiCo in the marketplace (see exhibits 5 and 6 for examples).

7 Kantar was a company that provided insights to brands to help them understand their perception by retailers. Kantar website, https://www.kantarmedia.com/uk.

Exhibit 4: PEPWORX℠



PEPWORX℠ was a PepsiCo-developed suite of data, tools, and processes created to help deliver more granular recommendations to retailers on product assortment, merchandising, and promotional activities.

Source: PepsiCo Demand Accelerator, December 2019.

PEPWORX℠ helped drive a variety of benefits for partners, from faster product turns and better profit realization to higher net price realization, because there was less of a need to discount products that weren't selling. On the consumer side, shoppers enjoyed tailored access to valued products.

Our work is not altruistic, but when we have the goal of serving the shopper and serving the customer—and we are genuine about that—and we have the portfolio and the capability to deliver against the opportunities when you find them, then we're going to win.

JEFF SWEARINGEN, SENIOR VICE PRESIDENT, PEPSICO DEMAND ACCELERATOR

Exhibit 5: Brand Accelerator Use Case

BRAND ACCELERATOR: CHEETOS ASTEROIDS

Cheetos Asteroids had been discontinued in the early 2000s, but people were discussing the product on social media and making requests for it. Some even signed a petition. Although PepsiCo marketers appreciated the product excitement, they considered the appeal narrow. To justify relaunching Asteroids, they would need to know just how to reach the niche audience of shoppers.

As a first step, the DX Media and Data team used social media data to understand where Cheetos Asteroids conversations were happening. Then they overlaid the locations with shopper data to identify convenience stores that were most likely to serve people who are asking for the product. Next, the team invested heavily in messaging and placed product in those locations. They viewed the results as quite successful:

Cheetos Asteroids did far better than our expectation. We could not keep them in stock. It was a combination of pent-up demand and love for this product—and also a very precise approach. We put the product in just the right locations, which made everybody really want it.

JENNIFER SAENZ, PRESIDENT, PEPSICO GLOBAL FOOD

Exhibit 6: Space and Optimization Use Case

SPACE AND OPTIMIZATION: PLANOGRAMS

DX leaders believed that optimizing store-level assortment was one of the easiest ways to drive revenue growth, at both the category level and for PepsiCo. Typically, retailers assorted their stores one way across the country, or possibly across a region. DX associates learned that assortment decisions should be made more granularly, because a store in urban Denver, for example, could have a shopper profile more like a store in urban Phoenix than with a suburban Denver store just fifteen minutes away.

The DX Space Optimization and Transformation team created an approach to tag individual stores with an identifier that reflected their local shopper base. Then, the team created planograms[8] for about seven distinct store profiles for a retailer. When DX had data about a retailer's total category, the team helped the retailer allocate assortment across all categories in the store, not just the categories in which PepsiCo operated.

In many cases, we're advising retailers on the entire category. Can shoppers find what they want? Can they find other commonly co-purchased items? Can they find new items?

ANKUR MEGHANI, SENIOR DIRECTOR, SPACE OPTIMIZATION AND TRANSFORMATION, PEPSICO DEMAND ACCELERATOR

With the launch of PEPWORX℠, DX associates believed that they had established capabilities that were making valuable impact for PepsiCo.

I'm proud of how we leverage insights like store-level clustering—and how those insights have become the language of the business. The concept of granularity is in more conversations than you can imagine. Then personally on my team, when we do the work the right way, the shopper wins, the customer wins, [the] company wins. And so in that way, too, I am very proud.

ANKUR MEGHANI

DX associates documented and shared successful solutions. They spoke to internal groups at forums and on webinars about technical concepts like machine learning and artificial intelligence to educate a wider audience of PepsiCo associates. The team created a DX portal that contained learning materials including ready-to-use content, successful use cases, online training, videos, and demonstrations. File downloads often averaged 10,000–15,000 per month, with seventy-five percent of hits coming from beyond the DX team.

The DX also launched a DX University, which explained the essence of each DX team, and provided varying degrees of training on each teams' capabilities. For example, a course on Shopper Analytics and Insights offered basic training on shopper behavior, while more in-depth courses were available for people interested in learning even more about the topic. The curriculum was delivered via the PepsiCo University learning portal so that anyone at PepsiCo could learn about DX topics.

The DX grew its data and insights beyond the United States to include Canada. And the European and Latin American sectors began to replicate the DX in their own markets and provide similar capabilities. Collectively, these teams shared best practices and informally operated as a global unit.

A DRIVE TOWARD EVEN DEEPER PARTNERSHIPS

In 2019, the DX was reaching its fifth anniversary. In a reference to the "Day One" mentality that Jeff Bezos created at Amazon,[9] DX leaders called for a "Year One" mentality to encourage new creative and aggressive future

8 A planogram is a diagram that indicates the prescribed placement of retail products on shelves to drive maximum sales results.

9 Brian Roemmele, "What Is Jeff Bezos's "Day 1" Philosophy?" Quora response published by Forbes, April 21, 2017; https://www.forbes.com/sites/quora/2017/04/21/what-is-jeff-bezos-day-1-philosophy/#158784771052; original response on Quora, https://www.quora.com/What-is-Jeff-Bezos-%E2%80%9CDay-1%E2%80%9D-philosophy.

goals. The leaders did not want their associates to become complacent with success. They wanted the unit to act more like a scrappy startup than an established part of the PepsiCo operating model.

One Year One goal was to expand the influence of shopper insights—and to find ways for PepsiCo to execute on the DX insights in more proactive and automated ways. The DX team identified and created shopper demand, whereas the Customer Supply Chain and Go-To-Market units executed that demand at the store level. Thus, DX leaders saw great opportunity in working with the Customer Supply Chain and Go-To-Market team to identify ways that shopper insights could influence PepsiCo's end-to-end supply chain. Customer Supply Chain had spent several years building data-based capabilities such as operational dashboards and frontline sales reporting and alerting tools to support the unit's field activities (see exhibit 7).

Exhibit 7: Select Customer Supply Chain Capabilities

Demand Signal Repository (DSR): The Demand Signal Repository was a global data repository that contained granular daily scan data from retailers around the world. The data was standardized and stored on a secure, cloud-based platform, and it was made available in a highly controlled manner, based on specific, required permissions. This scan data served as the foundation for a number of field execution tools, including the following four cloud-based tools:

1. ***FAST*** was a mobile front-line sales reporting and alerting software application. The application helped sales associates identify and close store-level execution gaps. Frontline associates automatically received alerts regarding the stores they were responsible for servicing via their mobile devices.
2. The **Innovation Tracker** was a dashboard application that helped sales representatives and managers track the distribution of individual SKUs. The application followed product distribution and performance.
3. **PromoX** was a software application that field teams used to monitor national or regional promotions as they were executed at the store level. The application drew upon daily scan data, store receipts, and shipment data, and indicated whether the right SKU-level inventory had been placed in the designated store on the targeted day.
4. Customer Supply Chain built a **test and learn dashboard** that the unit's associates used to monitor and measure various treatments (e.g., the introduction of a new SKU, the use of a new promotional material). The dashboard helped users test and learn whether activities were delivering the expected outcomes.

Collaborating with our retail partners to capture, cleanse, and harmonize daily scan data at the SKU level has enabled us to build a comprehensive set of innovative retail execution tools. These efforts have created a connected supply chain with our customers that allows us to partner to drive store-level execution and topline sales.

JOHN S. PHILIPS, SENIOR VICE PRESIDENT, PEPSICO CUSTOMER SUPPLY CHAIN AND GO-TO-MARKET

DX leaders believed that by increasingly integrating PepsiCo supply-side and demand-side capabilities, they could help position PepsiCo to deliver shopper insights—and action—to the company's retail partners. This connection between demand plans and shelf inventory was expected to more greatly deepen PepsiCo's customer partnership efforts.

Moving forward, the PepsiCo Demand Accelerator was expected to continue to impact PepsiCo in new, important ways. And, PepsiCo leaders believed that the DX held great promise for making increasingly local impacts based on granular insights.

What I appreciate about the DX beyond what it has delivered so far for PepsiCo is its future potential, especially for our beverage business where we aim to be nationally great, locally even better. This vision means we continue to drive for excellence at the national level building the strongest brands in the business while enabling greater customization and agility at the local level to enhance our competitiveness. The DX impact will continue to grow as we connect more directly with our regional divisions to tackle local opportunities, which will lead to stronger retail customer relationships and overall growth for PepsiCo.

KIRK TANNER, CEO, PEPSICO BEVERAGE NORTH AMERICA

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